HOSPITAL PERCEPTION POST COVID-19

Objective: This project aimed to investigate how patient trust in U.S. hospitals changed before and during the COVID-19 pandemic. Respondents are asked to assess hospitals across multiple measures, including.

- Communication (1) Communication with Doctors, (2) Communication with Nurses, (3) Responsiveness of Staff
- Hospital Environment: (1) Cleanliness, (2) Quietness
- Discharge Protocol: (1) Care Transition, (2) Discharge Information.

The primary metric used for this assessment was the Net Promoter Score (NPS), a widely recognized indicator of consumer loyalty and trust across various industries.

Data Source: The data was obtained from the HCAHPS survey, which assesses patient perspectives on hospital care from the year 2015 to 2023.

Thought Process

Why NPS?

NPS is the difference between the top percentage scores (Promoters) and the bottom percentage scores (Detractors), hence it considers the opinions of both Advocators and Opposers. Basically, NPS is used to measure the overall satisfaction of members towards hospital care.

Why focus on the pandemic?

COVID-19 brought about challenges to healthcare sector, especially hospitals where new safety protocols and social distancing measures were introduced. Furthermore, due to doctors, nurses and staff's increased workload that came with the uptick in patient admission, communication and responsiveness were compromised. These factors affect patients' perception towards hospital care negatively. Therefore, the purpose of this project is to understand areas that experienced the biggest decline and come up with plans to address those issues to regain patient trust.

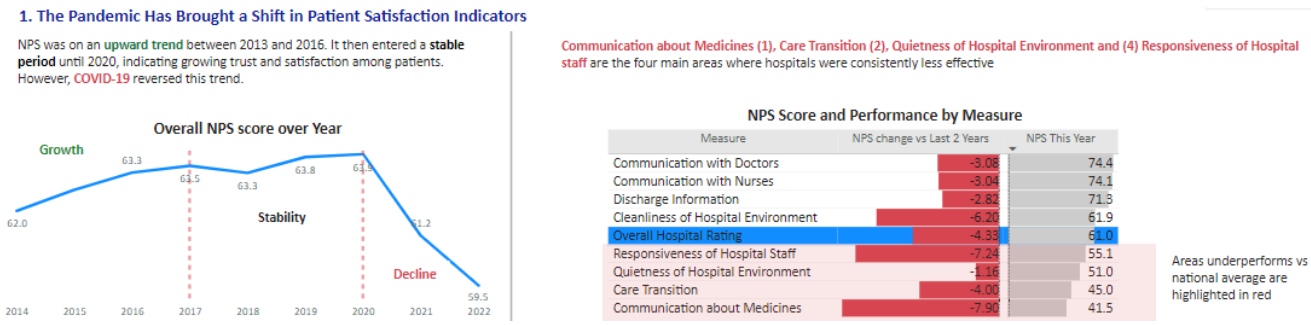
How is the project structured?

The project includes two main parts, namely Executive Summary and State Analysis. The Executive Summary part contains most salient findings from the data, including NPS trend and NPS performance by measure and by state. It concludes with key recommendations on how to tackle those weaknesses. The State Analysis part drills into the current state of Texas hospitals in terms of patient satisfaction. It analysis key drivers that contribute to their success as well as areas for improvement to boost their NPS score.

Each part starts from high level and goes into depth based on findings from the preceding sections

A- PART 1: EXECUTIVE SUMMARY

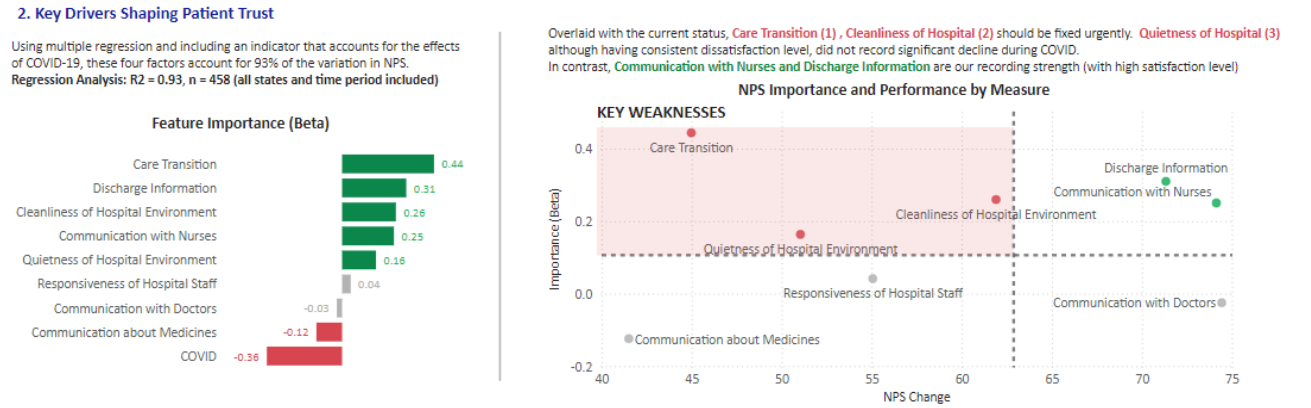
1. Section 1: The pandemic's impact: A shift in patient satisfaction indicators



This section starts with historical trend of NPS between the year 2014 and 2022 (line chart on the left). NPS recorded a growth phase from 2014 to 2017, then entered a stable period over the next four years until COVID-19 hit. NPS plummeted to 59.5, marking the lowest point since the survey started.

The section on the right shows the breakdown of NPS performs by different metrics. Four key areas that have lower score than the overall NPS were highlighted as they pose the biggest challenges to hospitals. They are (1) Responsiveness of Hospital Staff, (2) Quietness of Hospital Environment, (3) Care Transition and (4) Communication about Medicines.

2. Section 2: Identify key drivers shaping patient trust

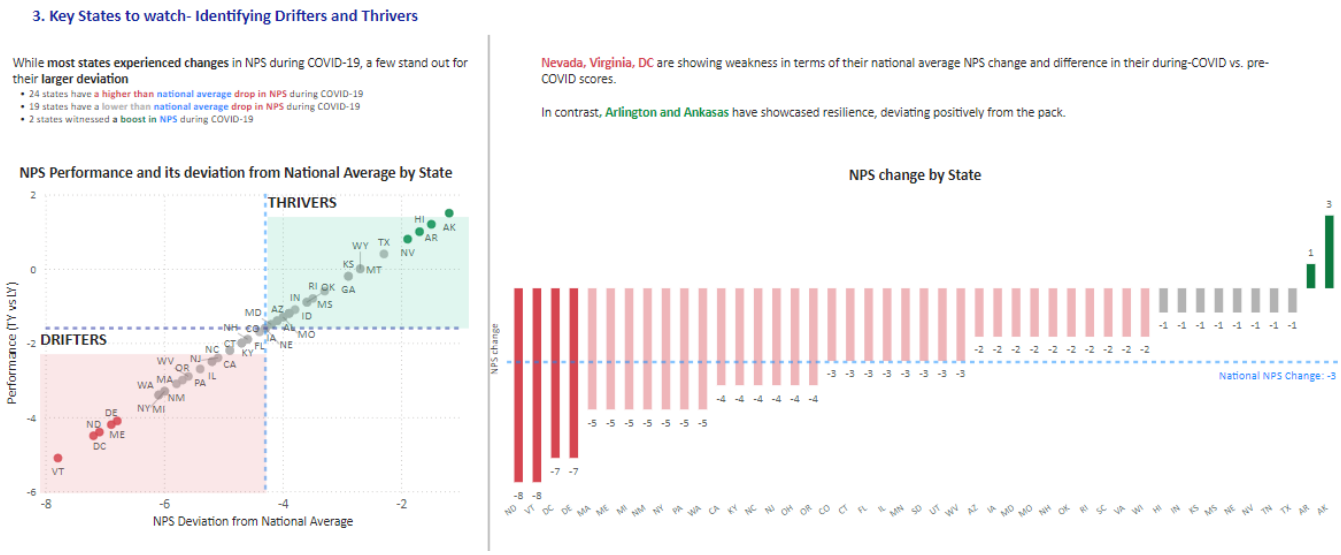


In this section, I am going to identify the most important factors that affect NPS. I used a multivariate regression analysis to test their significance against the NPS as the dependent variable. I used the state data and transformed it so that I could perform the regression analysis, resulting in 458 unique state-year observations. In order to test if the pandemic had an impact, I also included a dummy variable where =1 represents the COVID-19 era and =0 represents the pre COVID-19 era in the model. With an accuracy of 0.93, I can conclude that the model is statistically significant in identifying metrics that have highest influence on NPS.

The model shows that COVID-19 has a had a negative impact on NPS, reducing it by 0.36 point. (1) Care Transition, (2) Discharge Information, (3) Cleanliness of Hospital Environment, (4) Communication with Nurses and (5) Quietness of Hospital significantly elevate patient trust, increase NPS by >0.16 point for every unit increases.

In order to narrow on the key weaknesses that hospitals face, the chart on the right overlays importance with current NPS performance. (1) Care Transition, (2) Cleanliness of Hospitals, (3) Quietness of Hospital Environment, despite being among the most important measures that shape NPS, recorded the worst performance.

3. Section 3: Understand key states to monitor- Identify Drifters and Thrivers



In this section I am going to identify states that perform exceptionally well and unwell. I calculated the average (national) decrease in NPS and used it as a "benchmark" to identify which states are drifters and thrivers. **Drifters** are the states where there was a notable negative deviation in NPS scores after the onset of the COVID-19 pandemic. **Thrivers** on the other hand are states that despite the challenges brought by the pandemic managed to either maintain their previous NPS scores or even improve them. By combining the deviation from national average and the difference in NPS during vs pre pandemic, I was able to identify states that notably diverged from the "norm".

4. Section 4: Roadmap to rebuilding patient trust

The last section wraps up the project by summarizing actions plan that address challenges I have identified.

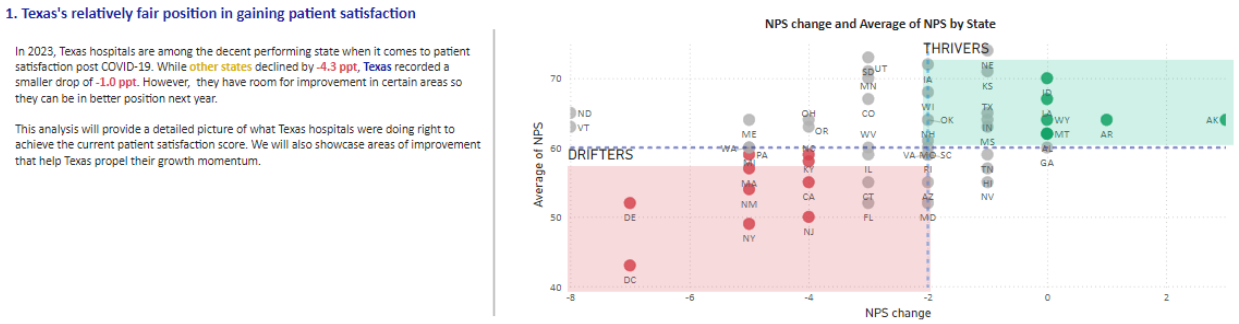
4. Recommendations to gain patient trust

No.	Key Challenges	Issues	Solutions
1	Widespread decline in trust	COVID-19 brings about an unprecedented changes in hospitals, including altered care procedures and change in hospital environment. This shift might not always have aligned with patient satisfaction, leading to a drop in NPI	Prioritize transparent communication about changes in care procedures and safety protocols. Investigate best practices in Arlington and Arkansas to understand areas they did well so we can replicate in our states.
2	Care Transition & Discharge Info: Critical in shaping patient trust but underperforms	Despite its strong link with NPI, it consistently scores lower than other measures	Implements post-discharge communication protocols. This will help clear patient confusion upon discharge.
3	Cleanliness of Hospital Environment: Important but neglected during COVID time	Due to staff shortage, hospital environment was not in the best condition.	Establish a proper hospital cleaning protocols to ensure staff do not miss any key areas or points of contact during regular cleaning rounds.
4	Inconsistent feedback collection timing	Since feedback collection window is big (1-10 days), patient's recollection may vary. More quality feedback could be provided when patients' experience is fresh	Narrow down feedback collection window to 1-3 working days

B- PART 2: STATE ANALYSIS

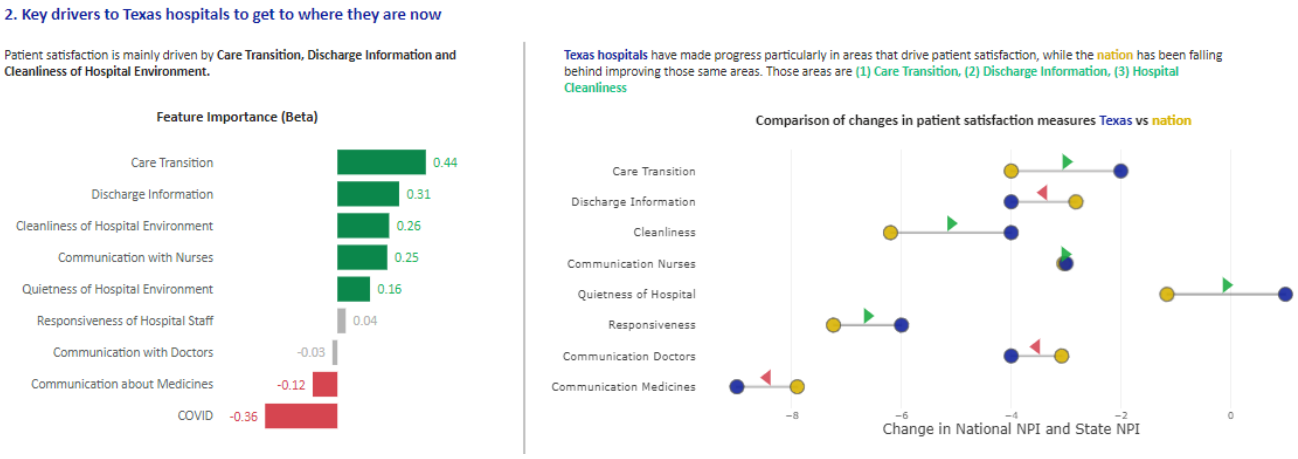
I will compare Texas and other states/ national average to understand areas they did well/ lacking. Texas is highlighted in blue and national data is in yellow for the ease of follow up.

1. Section 1: Texas's decent position in patient satisfaction



This section shows states in terms of their current NPS score and performance compared to 2 years ago. While other states declined by 4.3 ppt, Texas recorded a smaller decrease of 1.0 ppt.

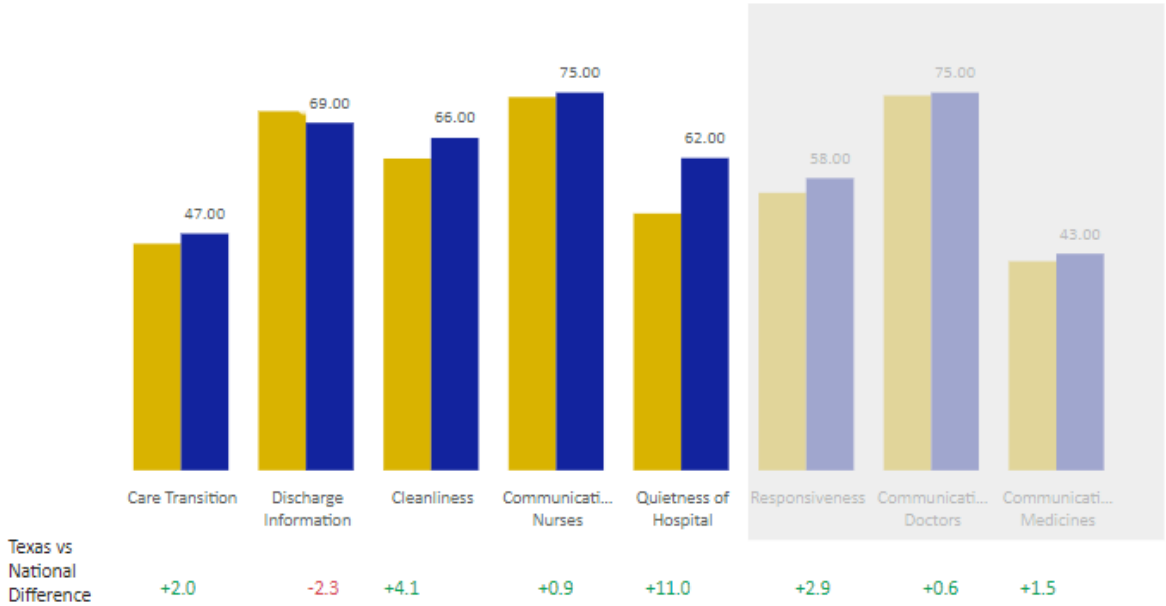
2. Section 2: Identify key drivers that contribute to the performance of Texas in patient satisfaction



The chart on the left displays most important features in shaping patient satisfaction. I will then proceed to show how Texas performs in those areas in relation to the nation. The reason Texas hospitals did relatively better than national average is the former made progress in areas that drive satisfaction while the nation fall behind in improving those same areas.

3. Section 3: Identify areas that Texas hospitals still can work on to improve their NPS further.

Not all areas with lower satisfaction scores are concerning since many of them do not have a strong impact on overall patient satisfaction. It is not about fixing everything, but about strategically enhancing what matters most to patients. Discharge information, despite it being critical in shaping patient satisfaction, is lacking compared to national average



4. Section 4: Texas hospitals' action plans for winning at patient satisfaction.

This part concludes the State Analysis section where I come up with key recommendations on areas what Texas hospitals can fix to boost their NPS in the upcoming year.